Course Syllabus

Edit

Introduction to Statistics and Data Science

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Quick Links

Evaluation

Late Work Policy

AI Policy

Course Description

This course introduces students to the discipline of statistics as a science of understanding and analyzing data. Students will learn the importance of data collection and sampling, methods to analyze data, and how to use data to make inferences and conclusions about real world phenomena. Students will be introduced to the free statistical programming software, Posit Cloud, to apply both descriptive and inferential statistics to real data sets.

Course Learning Objectives

- Use statistical software to manage and process data.

- Use statistical software to perform exploratory data analyses. That is, explore data numerically and visually to gain understanding through data and generate hypotheses and inferences to later test.
- Recognize the importance of data collection, identify limitations in data collection methods, and determine how they affect the scope of inference.
- Build a conceptual understanding of the unified nature of statistical inference.
- Apply estimation and testing methods to analyze single variables or the relationship between two variables in order to understand natural phenomena and make data-based decisions.
- Model numerical response variables using a single or multiple explanatory variables.
- Interpret results in context without relying on statistical jargon.
- Critique and evaluate data-based claims and decisions.

Course Materials

Textbook

We will be using **Introduction to Statistics and Data Science** **(https://nustat.github.io/intro-stat-data-sci/)** which is a free online book that we have been developing for this course.

Software

We will be using/introducing the free statistical software R with RStudio accessed through **Posit Cloud** **(https://posit.cloud/)**. Generous donations from the Provost's Office, WCAS Dean's Office, and Statistics and Data Science faculty members cover Posit Cloud student fees for this course.

Hardware

Students will need a laptop or Chromebook to be able to follow lectures and to work with Posit Cloud to complete activities.

If access to a laptop is an issue, then please contact the course instructor and we will work to find an accommodation. This requirement will not prevent students from taking this course.

Evaluation

Students will be evaluated through (1) small assignments; (2) reading tutorials; (3) homework assignments; (4) in-class learning checks; (5) 2 exams; and an (6) optional final exam.

- Small assignments: miscellaneous assignments such as surveys and short reading responses.
- Reading tutorials: completed using "Tutorials" on Posit Cloud. Each reading tutorial will be scaled to be worth 10 points and uploaded to the course Canvas page.
- Homework assignments: there will be roughly 6 homework assignments.
- In-class learning checks: during class there will typically be a short question(s) related to the lecture and/or reading. You can collaborate and ask questions to complete the learning check. These must

be completed during the window it is available during class.

- 2 exams: each exam will consist of two parts (1) coding/application focused (2) conceptual focused. Both exams are in-class. You are permitted 1 page (8.5x11) of notes front and back, any other resource is prohibited. The exams are not cumulative.
- Optional final exam: cumulative exam structured similarly to the exams. More detailed information is provided in the Assessment and Grading section below.

Assessment and Grading

Table provides the assessment categories, their weights, and the grading scale for the course.

Category	Weight		Grading Scale	
Small Assignments	5%		93.0- 100%	A
Reading Tutorials	5%		90.0 - 92.9%	A-
Learning Checks	10%		87.0 - 89.9%	B+
Homework	20%		83.0 - 86.9%	B
Exam 1	30%		80.0 - 82.9%	B-
Exam 2	30%		77.0 - 79.9%	C+
			73.0 - 76.9%	C
			70.0 - 72.9%	C-
			60.0 - 69.9%	D
			Below 59.9%	F

Optional Final Exam Policy

We have worked to develop a policy geared towards a growth mindset. That is, we want a policy where students clearly demonstrate that they have used exams as learning tools to improve their understanding of statistics. Therefore, we have an optional final exam which is cumulative and can **only improve** your grade.

Your final grade will be the **BETTER** of the following two calculation.

Category	Weight
Small Assignments	5%
Reading Tutorials	5%
Learning Checks	10%
Homework	20%
Exam 1	30%
Exam 2	30%

Category	Weight
Small Assignments	5%
Reading Tutorials	5%
Learning Checks	10%
Homework	20%
Exam 1	15%
Exam 2	15%
Final Exam	30%

Late Work Policy

We have worked to craft a late work policy that allows reasonable flexibility while supporting student learning by keeping them on schedule. Each assessment type has a late work policy that is tailored to its design and objectives. Students should expect the late work policy to be strictly enforced as flexibility has been designed into it.

Small Assignments & Reading Tutorials

- For each assessment type, students may submit up to 4 assignments late within 48 hours of the due date without penalty. Any additional late submissions will receive a score of 0.
- No assignments will be accepted more than 48 hours after the due date without prior approval.

In-class Learning Checks

- No late submissions! These must be completed during the window it is available during class.
- The 3 lowest learning checks scores will be dropped.

Homework Assignments

- Any late submission within 48 hours of the assignment due date will incur a 10% late penalty.
- No assignments will be accepted more than 48 hours after the due date without prior approval.
- The lowest homework score will be dropped.

Exams

- No late submissions! These must be completed in-class on their assigned dates.

Course Policies & Resources

Northwestern University Syllabus Standards

This course follows the **[Northwestern University Syllabus Standards (https://www.registrar.northwestern.edu/registration-graduation/northwestern-university-syllabus-standards.html)](https://www.registrar.northwestern.edu/registration-graduation/northwestern-university-syllabus-standards.html)** . Students are responsible for familiarizing themselves with this information

Attendance & Participation

While we do not collect formal attendance, implicit in the course design it is expected that you attend class to benefit from working with others – either by others helping you or by you helping others learn.

Use of Generative AI (for example ChatGPT)

In this course, generative AI resources are only to be used **with the purpose of supporting their learning.** Only turn to generative AI as a last resort/step. Never begin with it. Without developing a basic understanding first, you won't be able determine the usefulness or correctness of its output. To ensure academic integrity, students must openly disclose any AI-generated material they utilize and provide proper attribution. This includes in-text/code citations, quotations, and references. **Generative AI cannot be used on any exam.**

When using these tools students will typically be asked to create personal accounts for artificial intelligence services and/or software. Students should familiarize themselves with the Terms of Use for these services as well as the expectations around data privacy and use. Students **should not share** private, or otherwise sensitive information or data, about themselves or others, with these tools, as there is often no guarantee of data privacy.

Student Support Services

Student Enrichment Services (SES) partners with FGLI students – pronounced figly – who are first-generation, lower-income, and/or DACA/Undocumented. SES works with these students to foster identity development, navigate campus resources, and build community. Through campus-wide partnerships and advocacy, SES strives to build an inclusive Northwestern community that is welcoming, supportive, and accessible for all students. SES can connect you to a number of resources (emergency aid assistance, books, supplies, laptops, food accessibility and much more!) both within the SES office and across campus. For more information please visit **https://www.northwestern.edu/enrichment (https://www.northwestern.edu/enrichment/)** .

Academic Resources

Quarter-Long Study Groups (https://www.northwestern.edu/academic-support-learning/course-support/peer-guided-study-groups/) offers peer-led academic support in a small-group setting for students enrolled in this course. If you join the program, you will meet weekly with about 5 to 8 other students and a peer facilitator. During sessions, students review concepts, work through practice problems, raise questions, and work together to develop answers. Students register for the full quarter on CAESAR and weekly attendance is expected. Study Groups sessions are listed below course lecture and discussion sections (e.g., STAT 202-SG Peer-Guided Study Group: Introduction to Statistics and Data Science).

Drop-In Peer Tutoring (https://www.northwestern.edu/academic-support-learning/course-support/drop-in-peer-tutoring/) (No Appointment Needed) is set up such that students can drop in to study alone or with others and ask questions of a peer leader who has done well in the class. Tutoring is provided for many of the introductory courses in Biology, Chemistry, Economics, Engineering, Mathematics, Physics, and **Statistics**. Check their website for a complete list of supported courses.